Practice Quiz #3 Name: NetId:

1. Suppose we have conditional exchangeability given some covariate $L$. In the table below, $A_i$ denotes the observed treatment, $Y_i$ denotes the observed outcome, $\hat{E}(Y \mid A = 1, L)$ denotes the predicted outcome under treatment,$\hat{E}(Y \mid A = 0, L)$ denotes the predicted outcome under no treatment, and $\hat{\pi}_i = \hat{Pr}(A \mid L = \ell_i)$ denotes the predicted propensity score.

| Individual | $A_i$ | $Y_i$ | $\hat{E}(Y \mid A = 1, L)$ | $\hat{E}(Y \mid A = 0, L = \ell_i)$ | $\hat{\pi}_i$ |
|---|---|---|---|---|---|
| 1 | 1 | 210 | 180 | 100 | .5 |
| 2 | 1 | 160 | 180 | 100 | .8 |
| 3 | 0 | 60 | 80 | 60 | .2 |
| 4 | 0 | 70 | 80 | 60 | .3 |

(a) Which choice below calculates an estimated **average treatment effect on the treated (ATT)** via the g-formula using the individuals in the table?

◯ $\frac{1}{2}(180 + 180) - \frac{1}{2}(100 + 100)$
◯ $\frac{1}{4}(210 + 160 + 60 + 70) - \frac{1}{4}(180 + 180 + 80 + 80)$
◯ $\frac{1}{4}(180 + 180 + 80 + 80) - \frac{1}{4}(100 + 100 + 60 + 60)$
◯ $\frac{1}{4}(210 + 210 + 60 + 70 + 70) - \frac{1}{4}(180 + 180 + 80 + 80)$
◯ $\frac{1}{2}(80 + 80) - \frac{1}{2}(60 + 60)$

(b) Which choice below calculates an estimated **average treatment effect (ATE)** via the g-formula using the individuals in the table?

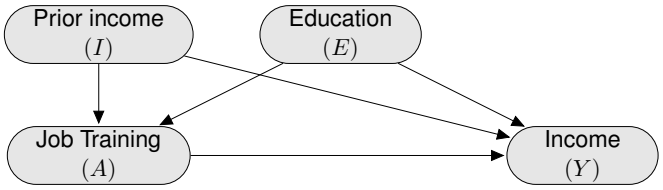
◯ $\frac{1}{2}(180 + 180) - \frac{1}{2}(100 + 100)$
◯ $\frac{1}{4}(210 + 160 + 60 + 70) - \frac{1}{4}(180 + 180 + 80 + 80)$
◯ $\frac{1}{4}(180 + 180 + 80 + 80) - \frac{1}{4}(100 + 100 + 60 + 60)$
◯ $\frac{1}{4}(210 + 210 + 60 + 70 + 70) - \frac{1}{4}(180 + 180 + 80 + 80)$
◯ $\frac{1}{2}(80 + 80) - \frac{1}{2}(60 + 60)$

(c) Which choice below calculates an estimated **average treatment effect (ATE)** via IPW using the individuals in the table?

◯ $\frac{1}{4}[(210/.5 + 160/.8) - (60/(1 - .2) - 70/(1 - .3)]$
◯ $\frac{1}{4}[(210/.5 + 160/.8) - (60/.2 - 70/.3)]$
◯ $\frac{1}{4}[(210/(1 - .5) + 160/(1.8) - (60/(1 - .2) + 70/(1 - .3)]$
◯ $\frac{1}{4}[(210/(1 - .5) + 160/(1 - .8) - 60/.2 + 70/.3]$

2. Suppose you are trying to build a model to predict $Y$ given $A$ (the treatment) and $L$ some covariate. You have the option of fitting a linear regression or using a very complex machine learning model. Suppose you only have 30 data points, which model would you prefer to use? Suppose you only have 30,000 data points, which model would you prefer to use? Give a short explanation of why (1-2 sentences).

3. Consider the causal graph below where we are interested in the causal effect of hours studied on exam scores. Note that $\{C, E\}$ is a sufficient adjustment set.




   (a) Suppose we know that $E(Y \mid A, I, E)$ is linear, but $P(A \mid I, E)$ is a very complex function. Which of the following methods would be a more reliable choice for estimating the ATE

   ◯ IPW ◯ Standardization / G-formula

   (b) Suppose we know that $E(Y \mid A, I, E)$ is linear, but $P(A \mid I, E)$ is a very complex function. We fit a linear regression for both $P(A \mid M, E)$ and $E(Y \mid A, C, E)$. Which of the following methods will give a reliable estimate of the ATE? (may choose more than one)

   ◯ IPW ◯ Standardization / G-formula ◯ AIPW

   (c) Suppose we know that $E(Y \mid A, I, E)$ is linear, but $P(A \mid M, E)$ is a very complex function. We fit a linear regression for both $P(A \mid I)$ and $E(Y \mid A, I)$ and don't include education $(E)$ in either model. Which of the following methods will give a reliable estimate of the ATE?
   ◯ IPW ◯ Standardization / G-formula ◯ AIPW

4. Suppose you are using a matching procedure and are deciding between 1 to 1 matching and 5 to 1 matching.

   (a) Which choice would have higher bias?
   ◯ 1 to 1 ◯ 5 to 1

   (b) Which choice would have higher variance?
   ◯ 1 to 1 ◯ 5 to 1

5. Suppose you are matching on the following covariates: Current Income (measured in USD) and years of education (measured in years). Explain why using the Mahalanobis distance is a better choice when compared to Manhattan distance.

6. Suppose we have a more complex graph which requires matching on the 30 covariates. Each of the covariates might have varying levels of effect on probability of treatment. Explain why using propensity score matching might be preferred over coarsened exact matching.

7. What is the name of at least 1 other person in your group (not including you)